S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 11

Aim: Implement Linear Regression to predict continuous values using a given dataset.

Name of Student: P r a t h a m e s h D e s h m u k h

Roll No.: CS23115

Semester/Year: VI/III

Academic Session:

Date of Performance:

Date of Submission:

AIM: Implement Linear Regression to predict continuous values using a given dataset.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- To understand the concept of supervised learning.
- To understand Linear Regression and its importance in Machine Learning.
- To learn the relationship between dependent and independent variables.
- To implement Linear Regression using Python.
- To evaluate model performance using error metrics.
- To visualize regression results using graphs.

THEORY:

Introduction to Machine Learning

Machine Learning is a subset of Artificial Intelligence that enables systems to learn patterns from data and make predictions or decisions without explicit programming.

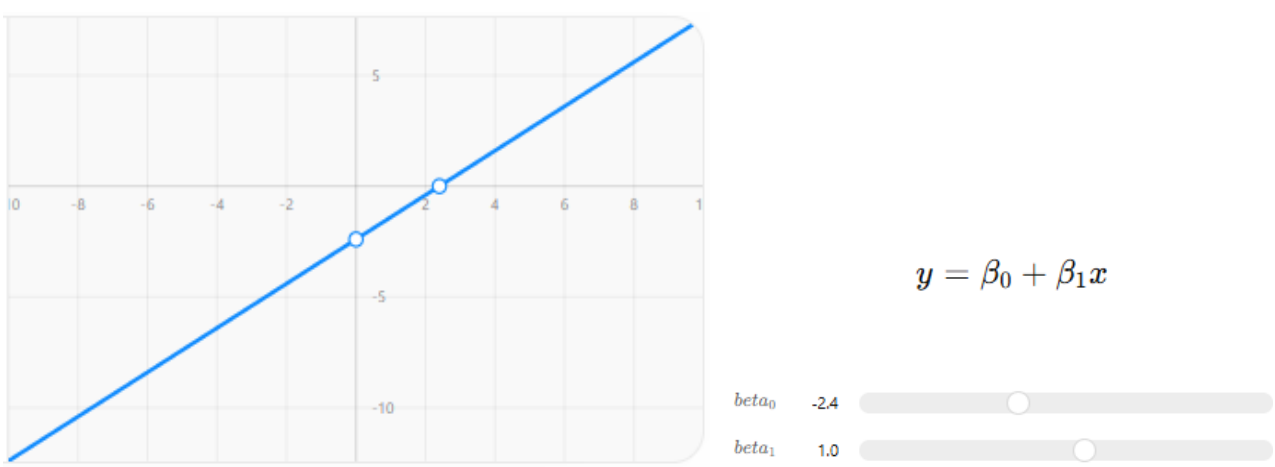
Supervised Learning

Supervised learning is a type of Machine Learning where the model is trained using labeled data. It includes:

- Regression (predict continuous values)
- Classification (predict categories)

Linear Regression

Linear Regression is one of the simplest and most widely used algorithms for predictive modeling. It establishes a linear relationship between input variables (X) and output variable (Y).



Where:

- y = Dependent variable
- x = Independent variable

- $\beta 0$ = Intercept
- $\beta 1$ = Slope (coefficient)

Types of Linear Regression

1. **Simple Linear Regression** – One independent variable
2. **Multiple Linear Regression** – More than one independent variable

Working of Linear Regression

- Collect dataset
- Plot data points
- Fit a best-fit line
- Minimize error using least squares method
- Predict new values

Error Metrics

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Advantages

- Simple and easy to implement
- Interpretable results
- Works well with linear relationships

Limitations

- Assumes linear relationship
- Sensitive to outliers
- Cannot capture complex patterns

ALGORITHM:

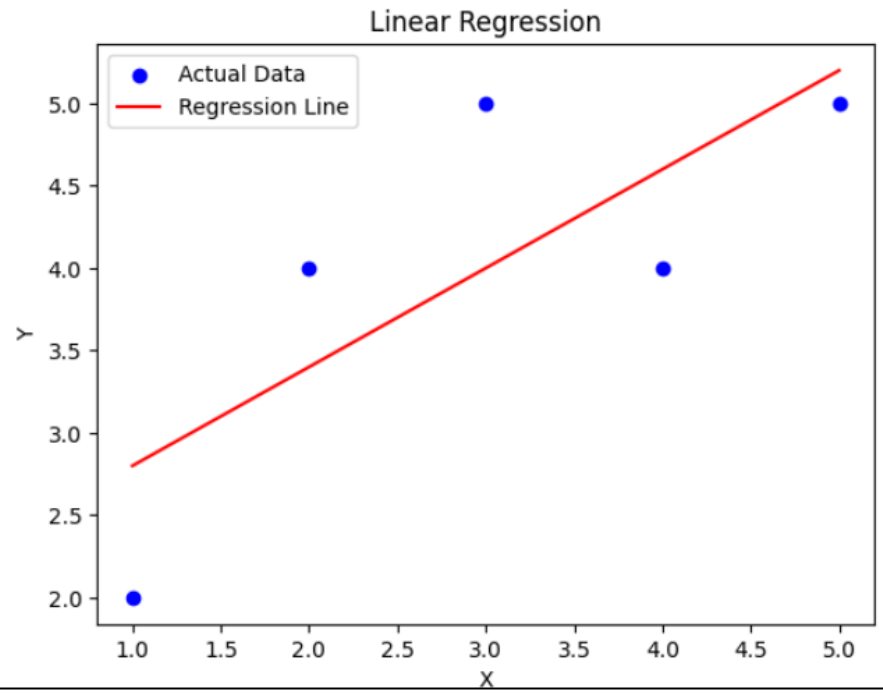
1. Import required libraries
2. Load dataset
3. Split dataset into training and testing sets
4. Create Linear Regression model
5. Train the model using training data
6. Predict output using test data
7. Evaluate model using error metrics
8. Visualize results

CODE:

```
# Import libraries
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score

# Sample Data
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
Y = np.array([2, 4, 5, 4, 5])

# Create Model
model = LinearRegression()

# Train Model
model.fit(X, Y)

# Predict
Y_pred = model.predict(X)

# Print model parameters
print("Slope (m):", model.coef_)
print("Intercept (c):", model.intercept_)

# Evaluation
print("Mean Absolute Error:", mean_absolute_error(Y, Y_pred))
print("R2 Score:", r2_score(Y, Y_pred))

# Plot graph
plt.scatter(X, Y, color='blue', label='Actual Data')
plt.plot(X, Y_pred, color='red', label='Regression Line')
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Linear Regression")
plt.legend()
plt.show()
```

INPUT & OUTPUT:

```
Slope (m): [0.6]
Intercept (c): 2.2
Mean Absolute Error: 0.6399999999999999
R2 Score: 0.6000000000000001
```



CONCLUSION:

DISCUSSION QUESTIONS:

1. What is Linear Regression?

2. Difference between simple and multiple regression?

3. What is the role of slope and intercept?

4. What are error metrics in regression?

5. What are limitations of Linear Regression?

REFERENCES:

- https://scikit-learn.org/stable/modules/linear_model.html
- https://www.geeksforgeeks.org/linear-regression-python-implementation/
- https://towardsdatascience.com/linear-regression-using-python